

CYBERSECURITY INCIDENT MANAGEMENT AND RISK ANALYSIS USING MYSQL

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Technology Used:	MySQL

1. Project Overview

Cybersecurity incidents such as malware attacks, unauthorized access, data breaches, and suspicious network activities can create serious risks for organizations.

This project focuses on analyzing cybersecurity incident data using MySQL. The main purpose of the project is to understand security incidents, identify high-risk organizations and systems, analyze login activities and network events, and find useful patterns from the available data.

SQL queries were used to perform filtering, grouping, aggregation, joins, subqueries, Common Table Expressions (CTEs), and window functions wherever required.

The analysis helps identify important areas that may require further security monitoring and investigation.

2. Problem Statement

Organizations generate a large amount of cybersecurity-related data through login activities, network events, security incidents, systems, and users. Simply storing this data is not enough. It is important to analyze the data and identify:

- Which types of security incidents are most common?
- Which incidents have the highest severity?
- Which organizations have the highest number of incidents?
- Which systems generate more incidents or network events?
- Which organizations have more failed login attempts?
- Which operating systems are used most frequently?
- Which areas may require additional security monitoring?

This project uses MySQL to answer these questions and generate meaningful insights from the cybersecurity dataset.

3. Objectives of the Project

1. To analyze cybersecurity incidents using MySQL.
2. To understand incidents based on severity and incident type.
3. To analyze security incidents across different years.
4. To analyze successful and failed login attempts.
5. To identify organizations with high security activity.
6. To identify systems with high numbers of incidents and network events.
7. To analyze operating system usage across systems.
8. To identify organizations with high numbers of Critical incidents.
9. To analyze network and authentication-related activities.
10. To derive useful security insights and recommendations from the data.

4. Dataset Description

The project database contains multiple tables related to cybersecurity management.

Main Tables Used	Purpose
organizations	Stores organization and industry information
users	Stores user-related information
systems	Stores information about organizational systems
login_logs	Stores login attempts and their status
network_events	Stores network-related activities
security_incidents	Stores cybersecurity incident details
incident_systems	Connects incidents with affected systems

The incident_systems table acts as a bridge between security incidents and systems.

5. Data Preparation and Database Setup

Before performing the main analysis, the cybersecurity database was prepared and validated in MySQL.

The database and its seven tables were reviewed using sample records, table structures, and record counts. Data types were modified where required, and date fields in the login logs, network events, and security incidents tables were converted into proper MySQL DATE format for analysis.

Data quality checks were performed to identify duplicate records and orphan records and to verify the relationships between the related tables.

Primary keys were then defined for the tables. The incident_systems table uses a composite primary key consisting of incident_id and system_id, as one incident can be associated with multiple systems.

Foreign key relationships were also created between the related tables, including:

- Users → Organizations

- Systems → Organizations
- Login Logs → Users
- Network Events → Systems
- Incident Systems → Security Incidents
- Incident Systems → Systems

After completing these preparation and validation steps, the database was used for the SQL-based cybersecurity analysis.

5. Tools and Technologies Used

MySQL was used as the primary database and analysis tool. The project used different SQL concepts including:

- | | |
|---|---------------------------|
| ▪ SELECT & WHERE Clause | ▪ GROUP BY & ORDER BY |
| ▪ Aggregate Functions (COUNT, SUM, AVG) | ▪ JOIN Operations |
| ▪ Subqueries & CTEs | ▪ Window Functions |
| ▪ CASE Statements | ▪ Percentage Calculations |
| ▪ Filtering and Ranking | |

These SQL concepts were used to extract meaningful information from the cybersecurity database.

6. Data Analysis and Key Findings

6.1 Incidents by Severity

The distribution of incidents by severity is presented below:

300#Q8.Find the number of security incidents for each severity level.
301 • select severity, count(incident_id) as total_incidents from security_incidents group by severity;
302

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	severity	total_incidents
▶	Medium	111
	Critical	140
	Low	129
	High	120

Severity	Number of Incidents
Critical	140
Low	129
High	120
Medium	111
Total	500

Insight

Critical incidents have the highest count with 140 incidents. This shows that a significant portion of the incidents in the dataset are classified as highly serious. Low-severity incidents are the second highest, followed by High and Medium incidents.

Conclusion

Critical incidents should receive higher priority during incident response because they can potentially have a greater impact on systems and organizational data.

6.2 Incidents by Year

Comparison of total cybersecurity incidents recorded across years:

303

#Q9. Find the number of security incidents discovered in each year.

304

•

select

year(discovered_date)

as

Year_Number,

count(incident_id)

as

total_incidents

from

security_incidents

group by

Year_Number;

305

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Year_Number	total_incidents
▶	2024	126
	2023	374

Year	Number of Incidents
2023	374
2024	126
Total	500

Insight

The dataset contains significantly more incidents in 2023 than in 2024. There were 374 incidents in 2023 compared with 126 incidents in 2024, which represents a large decrease.

Conclusion

The number of security incidents decreased significantly from **374 in 2023** to **126 in 2024**. This shows a clear change in incident volume between the two years. However, the incident count alone does not explain the reason for this decrease.

7. Login Analysis

7.1 Login Attempts by Status

Overall distribution of authentication attempts recorded in login_logs:

315 #Q12 Find the number of login attempts for each login status (successful/failed).
316 • select * from login_logs;
317 • select status, count(login_id) as login_attempts from login_logs group by status;
318

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	status	login_attempts
▶	Success	3040
	Failed	2960

Login Status	Attempts
Success	3,040
Failed	2,960
Total	6,000

Insight

The dataset contains 6,000 login attempts. Successful attempts are slightly higher than failed attempts. However, 2,960 login attempts failed, which is almost half of all login attempts.

Conclusion

The high number of failed login attempts indicates that authentication activity should be monitored carefully. Repeated failed attempts may require further investigation to identify possible account-access problems or suspicious login behavior.

8. Incident Type Analysis

8.1 Most Common Security Incident Type

```
581 # Q72. Find the most common security incident type in the database.
582 • select incident_type, count(*) as incident_count from security_incidents group by incident_type order by incident_count desc limit 1;
583
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
incident_type	incident_count			
Malware	185			

Most Common Overall: Malware – 185 incidents

Insight

Malware is the most frequently occurring incident type in the dataset.

Conclusion

The organization should give appropriate attention to endpoint protection, malware detection, system updates, antivirus/EDR monitoring, and employee security awareness.

8.2 Most Common Incident Type Among Critical Incidents

Most Common Critical Incident: Data Breach – 49 incidents

```
406 #Q34. Find the most common incident type among Critical-severity incidents.
407 • select incident_type, count(incident_id) as total_number, severity from security_incidents where severity = "Critical" group by
408 incident_type order by total_number desc limit 1;
409
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
incident_type	total_number	severity		
Data Breach	49	Critical		

Insight

Data Breach represents the largest category among Critical incidents.

Conclusion

Data protection should be an important security priority. Organizations should focus on access control, data protection, monitoring, and timely detection of suspicious activities.

9. Organization-Level Analysis

9.1 Organization with Highest High-Severity Incidents

ORG078 – 4 High-severity incidents

```
388 • select org_id, count(incident_id) as Number_of_incidents from security_incidents where severity = "High" group by org_id order by
389 Number_of_incidents desc limit 1;
```

org_id	Number_of_incidents
ORG078	4

Insight

ORG078 has the highest number of High-severity incidents in the dataset.

Conclusion

This organization can be considered for additional investigation to understand the causes and patterns behind its High-severity incidents.

9.2 Organization with Highest Critical Incidents

ORG058 – Healthcare – 5 Critical incidents

```
576 # Q71. Find the organization with the highest number of Critical incidents and display its industry.
577 • select o.org_id, o.industry, count(s.severity) as severity_count from organizations o join
578 security_incidents s using(org_id) where s.severity = 'Critical' group by o.org_id, o.industry order by severity_count desc limit 1;
```

org_id	industry	severity_count
ORG058	Healthcare	5

Insight

ORG058 has the highest number of Critical incidents, with 5 incidents. The organization belongs to the Healthcare industry.

Conclusion

ORG058 could be prioritized for further investigation because of its relatively high Critical incident count. The analysis can be extended to understand the incident types and systems associated with these Critical incidents.

9.3 Organization with Highest Total Incidents

ORG079 – Finance – 12 incidents

```
550 # Q65. Find the organization with the highest number of security incidents and display its industry.
551 • select o.org_id, o.industry, count(s.incident_id) as Highest_count from organizations o join
552 security_incidents s using(org_id) group by o.org_id, o.industry order by Highest_count desc limit 1;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
org_id	industry	Highest_count		
ORG079	Finance	12		

Insight

ORG079 has the highest total number of security incidents in the database.

Conclusion

ORG079 can be considered a high-priority organization for further analysis because it has the highest overall incident volume.

10. Network Event Analysis

10.1 Organization with Highest Network Events

ORG057 – Finance – 96 network events

```

588 # Q74. Find the organization with the highest number of network events and display its industry.
589 • select o.org_id, count(n.event_id) as network_events, o.industry from network_events n join systems s using(system_id)
590 join organizations o using (org_id) group by o.org_id, o.industry order by network_events desc limit 1;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
org_id	network_events	industry		
ORG057	96	Finance		

Insight

ORG057 has the highest number of network events in the dataset, with 96 events.

Conclusion

The organization may require closer network monitoring to understand why its network activity is higher than other organizations. However, a high number of network events does not automatically mean malicious activity. Further investigation would be required.

10.2 System with Highest Network Events

S0005 – 17 network events

```

467 # Q49. Find the system that has the highest number of network events.
468 • select system_id, count(event_id) as Event_count from network_events group by system_id
469 order by Event_count desc limit 1;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
system_id	Event_count			
S0005	17			

Insight

System S0005 has the highest number of network events among the systems analyzed.

Conclusion

S0005 could be selected for further investigation to understand the reason for its

higher network activity.

11. System Analysis

11.1 Operating System with the Highest Number of Systems

Windows – 213 systems

```
421 # Q38. Find the operating system type with the highest number of systems.
422 • with system_table as (select os_type, count(system_id) as system_count from systems group by os_type)
423 select os_type, system_count from system_table order by system_count desc limit 1;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
os_type	system_count		
Windows	213		

Insight

Windows is the most commonly used operating system in the dataset.

Conclusion

Since Windows represents the largest portion of systems, Windows-based systems should receive appropriate attention regarding security updates, patch management, endpoint protection, and vulnerability monitoring.

11.2 System with Highest Number of Incidents

S0031 – 4 incidents

```

451 # Q45. Find the system associated with the highest number of security incidents.
452 • select i.system_id, count(s.incident_id) as Incident_count from incident_systems i join
453 security_incidents s using(incident_id) group by i.system_id order by Incident_count desc limit 1;
454

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
system_id	Incident_count				
S0031	4				

Insight

System S0031 has the highest number of associated security incidents.

Conclusion

S0031 can be considered for further investigation to identify whether there are common vulnerabilities, repeated incidents, or other reasons for its higher incident count.

12. Failed Login Analysis

12.1 Organization with Highest Failed Login Attempts

ORG018 – Tech – 49 failed login attempts

```

495 # Q55. Find the organization with the highest number of failed login attempts.
496 • select u.org_id, count(l.status) as attempt_count from login_logs l join users u using(user_id) where status = 'Failed'
497 group by u.org_id order by attempt_count desc limit 1;
498

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
org_id	attempt_count				
ORG018	49				

Insight

ORG018 has the highest number of failed login attempts, with 49 failures. The organization belongs to the Tech industry.

Conclusion

The login activity of ORG018 should be monitored carefully. Repeated failed login attempts may be related to incorrect credentials, user-access issues, or potentially suspicious authentication activity. Further analysis of login timestamps, users, IP addresses, and repeated attempts would be useful.

13. Overall Key Insights

Based on the SQL analysis, the following major findings were identified:

1. Critical incidents are the highest severity category

Critical incidents account for the largest number of incidents, with 140 cases.

2. Malware is the most common incident

Malware has 185 incidents, making it the most common security incident type in the dataset.

3. Data Breach is a major Critical incident

Data Breach is the most common incident type among Critical incidents, with 49 cases.

4. Failed login activity is significant

There are 2,960 failed login attempts out of 6,000 total attempts, indicating that authentication activity requires monitoring.

5. Some organizations require closer attention

ORG058 has the highest number of Critical incidents, while ORG079 has the highest total incidents.

6. Network activity varies across organizations

ORG057 has the highest number of network events, with 96 events.

7. Windows is the dominant operating system

Windows is used by 213 systems, making it the most common operating system in the dataset.

8. Some systems have higher security activity

S0031 has the highest number of incidents, while S0005 has the highest number of network events.

14. Security Recommendations

Based on the analysis, the following recommendations can be made:

1. Strengthen Malware Protection

Since Malware is the most common incident type, organizations should focus on:

- Endpoint protection
- Antivirus/EDR monitoring
- Regular software updates
- System patching
- Employee security awareness

2. Improve Authentication Monitoring

Because a large number of login attempts failed, organizations should monitor:

- Repeated failed login attempts
- Suspicious login patterns
- Unusual access behavior
- Account lockouts
- Strong password policies
- Multi-factor authentication

3. Focus on Data Protection

Since Data Breach is the most common Critical incident type, organizations should strengthen:

- Access control
- Data encryption
- User permissions
- Data monitoring
- Backup and recovery procedures

4. Prioritize Organizations for Further Investigation

Organizations such as ORG058, ORG079, ORG018 and ORG057 can be prioritized for additional analysis based on their incident, login, or network activity.

5. Monitor High-Activity Systems

Systems such as S0031 and S0005 should be investigated further because of their higher incident or network-event activity.

6. Strengthen Windows Security

Since Windows is the most common operating system in the dataset, organizations should ensure regular:

- Security updates
- Patch management
- Vulnerability scanning
- Endpoint monitoring

15. Business Value of the Project

This project demonstrates how SQL can be used to convert raw cybersecurity data into useful information.

The analysis can help security teams:

- Identify high-risk incidents
- Monitor Critical incidents
- Detect unusual login activity
- Identify organizations requiring further investigation
- Monitor systems with high activity
- Understand common attack/incident types
- Prioritize security monitoring efforts

The project also demonstrates how database analysis can support data-driven cybersecurity decision-making.

16. Interpretation Note

The analysis identifies important patterns and potential risk indicators in the available cybersecurity data. These findings can support further investigation, but individual incidents, failed logins, or high activity levels should not be considered confirmed security threats without additional investigation.

17. Final Conclusion

The **Cybersecurity Incident Management and Risk Analysis Using MySQL** project demonstrates how SQL can be used to analyze cybersecurity data and identify important security patterns.

The analysis found that **Critical incidents** are the largest severity category, while **Malware** is the most common incident type overall. Among Critical incidents, **Data Breach** is the most common type, highlighting the importance of data protection and monitoring.

The login analysis showed **2,960 failed login attempts out of 6,000 total attempts**, indicating that authentication activity should be monitored carefully. At the organization level, **ORG058** had the highest number of Critical incidents, **ORG079** had the highest total incidents, **ORG018** had the highest failed login attempts, and **ORG057** had the highest network activity.

At the system level, **Windows** was the most commonly used operating system, while **S0031** had the highest number of associated incidents and **S0005** had the highest number of network events.

Overall, the project shows that SQL-based analysis can help identify important cybersecurity patterns, prioritize areas for further investigation, and support better data-driven cybersecurity decision-making.

18. Skills Demonstrated

Through this project, the following technical skills were demonstrated:

✓ MySQL	✓ SQL querying
✓ Data filtering	✓ Aggregation
✓ GROUP BY and ORDER BY	✓ JOIN operations
✓ Subqueries	✓ Common Table Expressions (CTEs)
✓ Window functions	✓ CASE statements
✓ Percentage calculations	✓ Data analysis
✓ Cybersecurity data interpretation	✓ Insight generation
✓ Problem solving	